

The Placeholder Force

How the Same F Conceals Different Physics

A companion to *Educational Phlogiston*, on constitutive laws, coupling constants, inertial mass, and the equation that hides them

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Educational Phlogiston argued that $F = ma$ gives no notational sign of which physical situation currently occupies it. Three scenarios press the point further. In each, a description of motion meets a separately supplied empirical law: magnetic coupling, elastic response, or gravitation. Newton's second law is the common hinge, but it does not say what has arrived at the hinge. In one case, and only one, the familiar notation also conceals an equivalence that took centuries to recognise.

NEWTON'S second law determines the acceleration once the net force is known; it does not determine what force is present, how it depends on position or velocity, or which property of the body produces the coupling. That missing information arrives from a separate empirical law, different in each of the three cases below.

Three scenarios, one hinge

A charged particle curving in a magnetic field, a mass oscillating on a spring, and a test body orbiting a central mass are rarely taught side by side. Yet each model has the same architecture: a kinematic description of the motion is joined to an empirical interaction or response law by

$$F_{\text{net}} = m_i a.$$

The equation supplies the bridge but does not supply the mechanism.

System	Kinematic nature	sig-	Empirical law	Couples via
Magnetic	$a = v^2/r$		$F_B = qvB$	charge q
Spring	$a = -\omega^2 x$		$F_s = -kx$	stiffness k
Orbit	$a = 4\pi^2 r/T^2$		$F_g = m_g^{(p)} g$	passive gravitational mass

In every row, inertial mass m_i appears in Newton's second law as resistance to acceleration. What changes is the law that gives F its value. Charge fixes the magnetic coupling; the spring's stiffness fixes its elastic response; passive gravitational mass fixes how strongly a body responds to a gravitational field. Only in the third case does experiment reveal a universal proportionality between the coupling property and inertial mass. The algebra alone does not provide it.

A charged particle in a field

Take a particle moving at constant speed in a circle in a uniform magnetic field, with its velocity perpendicular to the field. The geometry of the path gives $a = v^2/r$. The Lorentz force supplies the interaction, $F_B = qvB$. Newton's second law

connects them:

$$qvB = m_i \frac{v^2}{r} \implies r = \frac{m_i v}{qB}.$$

Charge decides how strongly the particle couples to the field; inertial mass decides how much the resulting force bends its path. Nothing requires the two properties to be related. Thomson's cathode-ray measurements isolated precisely such a charge-to-mass ratio [2], and mass spectrometers still exploit the same separation of roles.

A mass on a spring

A sinusoidal motion $x = A \cos(\omega t + \phi)$ has the kinematic signature $a = -\omega^2 x$, a statement about the form of the motion, not yet an explanation of why it occurs. The missing physics is the spring's constitutive response, $F_s = -kx$, valid within the linear elastic regime. This is literally a *constitutive law*: it describes how a particular material system responds to deformation, much as an *equation of state* like $pV = nRT$ supplies material information that conservation laws do not contain. Newton's second law transmits the response into an equation of motion,

$$m_i \ddot{x} = -kx \implies \omega = \sqrt{\frac{k}{m_i}}.$$

Here k is not a property of the moving body at all; it belongs to the spring, and more precisely to the spring within a specified deformation regime.

The Lorentz and indeed the gravitational laws are not constitutive in this strict material sense, being rather interaction laws. Yet they occupy the same logical position in their respective models providing the empirical closure that turns a general dynamical balance into a particular physical prediction.

Two kinds of mass

Orbital motion looks almost identical on the page to simple harmonic motion, and Bertrand's theorem [interactiveBertrand](#) tells us why¹

¹The inverse-square law and the linear restoring force are the only two central-force laws for which every bound orbit closes, a kinship deep enough that Newton proved both in successive propositions of the *Principia* (I,X and I,XI). That mathematical proximity makes it all the more important to see what is physically different.

For a circular orbit, $a = 4\pi^2 r / T^2$ is kinematics. To expose what the usual notation conceals, write the orbiting body's passive gravitational mass as $m_g^{(p)}$, its inertial mass as m_i , and the central body's active gravitational mass as $M_g^{(a)}$. Setting the gravitational force equal to $m_i a$,

$$\frac{G m_g^{(p)} M_g^{(a)}}{r^2} = m_i \left(\frac{4\pi^2 r}{T^2} \right) \implies \frac{r^3}{T^2} = \frac{G M_g^{(a)}}{4\pi^2} \frac{m_g^{(p)}}{m_i}.$$

The test body's mass disappears from the result only after an empirical fact has been supplied: the ratio $m_g^{(p)} / m_i$ is universal for all bodies, not merely for bodies whose mass is negligible. A conventional choice of units absorbs this common proportionality into G , allowing $m_g^{(p)}$ and m_i to be represented by the same symbol m [1]. With this normalisation,

$$\frac{r^3}{T^2} = \frac{GM}{4\pi^2}.$$

This is Kepler's third law, with the orbiting mass gone entirely, the same universality that makes a hammer and a feather fall together, now producing a regularity in the sky.

A note on two points that are easily conflated here. The *equivalence principle* is the claim that passive gravitational mass is proportional to inertial mass for every body; it applies regardless of the body's size.

The *test-body approximation* is the separate assumption that the smaller body's effect on the source can be neglected because $M \gg m$. In the exact Newtonian two-body problem, after the conventional identification of active gravitational, passive gravitational, and inertial mass for both bodies, the relative orbit obeys

$$\frac{r^3}{T^2} = \frac{G(M + m)}{4\pi^2}.$$

The additional m records the motion of the larger body, not a failure of equivalence.

“That $m_g^{(p)}$ cancels m_i is not an algebraic convenience but an empirical fact, elevated into the equivalence principle and absorbed by convention into G .”

Equation, definition, and closure

$F_{\text{net}} = m_i a$ connects an interaction model to a motion model. It can also serve operationally to measure inertial mass when force and acceleration are independently controlled. What it cannot do is identify the origin of the force or provide the empirical relation that gives the force its value. That missing relation may be a constitutive law, an interaction law, or a more elaborate field model. The orbital example then imports one further empirical statement: the gravitational coupling of matter is universally proportional to its inertial resistance. General relativity took that universality seriously enough to replace gravitational force with spacetime geometry, precisely because no other fundamental interaction couples universally to inertial mass-energy.

None of this is visible from the placeholder. $F_B = qvB$, $F_s = -kx$, and $F_g = G m_g^{(p)} M_g^{(a)} / r^2$ all occupy the left-hand side of the same balance.

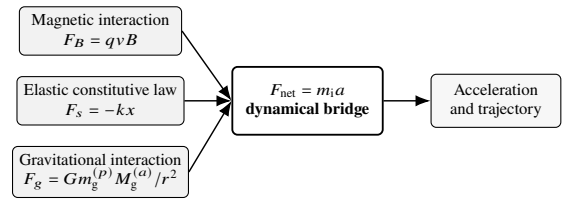


Figure 1: Different empirical laws feed the same dynamical bridge. The laws on the left supply the physical provenance of F ; Newton's second law transmits their consequence into motion but does not replace them.

Only a reader who knows the law that produced each expression can recover the physical mechanism, the relevant coupling, and the empirical assumptions that have been compressed into the symbols.

References

- [1] C. M. Will, “The confrontation between general relativity and experiment,” *Living Reviews in Relativity*, vol. 17, article 4, 2014. doi: [10.12942/lrr-2014-4](https://doi.org/10.12942/lrr-2014-4).
- [2] J. J. Thomson, “Cathode rays,” *Philosophical Magazine*, vol. 44, no. 269, pp. 293–316, 1897. doi: [10.1080/14786449708621070](https://doi.org/10.1080/14786449708621070).

The symbol F is not an explanation. Newton's second law is the meeting place between an interaction and a motion; only the empirical law supplying F tells us what kind of physics has arrived there.